

Rakesh Rosan Paital

Lead AI Engineer - Generative AI & Agentic Systems - WIPRO

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PROFILE SNAPSHOT

GenAI Lead Engineer with 8+ years of experience in Data Science, Machine Learning, Generative AI, and Agentic AI. Expertise in designing production-scale RAG systems, LLM applications, AI agents, and cloud-native AI solutions on GCP. Proven experience in deploying enterprise AI products, building intelligent automation systems, and leading end-to-end AI solution development from research to production.

SKILLS

AI & Machine Learning	Generative AI, Agentic AI, Machine Learning, Deep Learning, NLP
LLM & Prompt Engineering	Prompt Engineering, RAG, LangChain, LangGraph, Multi-Agent Systems, Guardrails, Vector Databases, Embedding Models
Deployment & Serving	FastAPI, Streamlit, Flask, Docker, Vertex AI
Programming & Scripting	Python, NupPy, Pandas
Cloud & DevOps	GCP, CI/CD
Data & Databases	MySQL, PostgreSQL

WORK EXPERIENCE

WIPRO(Standard Chartered Bank) May 2025 - continue
Gen AI Lead

1. Designed and implemented a production-ready Hybrid RAG architecture for structured audit data, reducing information retrieval time by 70% and improving answer relevance through row-level semantic retrieval. Leveraged row-level embeddings, vector search (top-k retrieval), and LLM-based response generation with follow-up query handling and feedback logging for continuous.
2. Deployed the solution on Databricks with Medallion architecture (Bronze/Silver/Gold), configuring Unity Catalog, vector search endpoints, LLM serving endpoints, and secure cluster policies for enterprise-scale usage.
3. Successfully implemented Server side logic and expose robust API's enabling the frontend team to interact efficiently with document generation service.
4. Contributed to enhancing document extraction accuracy by mitigating hallucination in outputs generated by OpenAI's LLM

CureBay Dec 2024 - May 2025
Data Scientist - mid

1. Spearheaded initiatives to improve data quality, governance, and compliance.
2. Implemented RFM clustering to classify customer segments, optimizing engagement strategies.
3. Designed and implemented intelligent ILM for task automation.
4. Designed a retrieval-augmented generation system to improve search efficiency.

IBM India Private Limited (Merative) March 2022 - Dec 2024
Senior Developer (Data Engineer)

1. Developed and deployed Generative AI-powered solutions to automate decision-making and improve business efficiency.
2. Developed an AI-powered system to streamline medical prescription generation.
3. Spearheading the development and implementation of predictive modeling, data processing, and data mining algorithms on Google Cloud Platform (GCP), optimizing

business decision-making processes.

4. Collaborated with cross-functional teams to translate business challenges into AI-driven solutions.
5. Lead data engineering initiatives, ensuring seamless data ingestion, transformation, and storage.
6. Created robust ML pipelines for data mining, classification, and clustering problems.
7. Designed and implemented intelligent AI agents for task automation, enhancing operational workflows.
8. Working in NLP, machine learning, and deep learning. Skilled in developing and fine-tuning models like GPT and BERT for conversational AI and automation.

BIPROS

Aug 2017 - Jan 2022

Analyst

1. Played a pivotal role in analyzing and processing unstructured data, employing advanced image processing and computer vision techniques for IMFA Pvt Ltd.
2. Deployed optimized computer vision models on edge devices, meeting, and exceeding customer requirements for real-time image analysis.
3. Engaged in direct client interactions, assessing technical requirements, and delivering tailored solutions that addressed specific operational challenges for **PowerGrid**.
4. Contributed to the design and implementation of database warehouses, optimizing data storage and retrieval processes for large-scale applications. Optimized SQL views, functions, and stored procedures, enhancing database performance and data quality.

EDUCATION

Master of Computer Application(Gandhi Institute for Technological Advancement, Bhubaneswar)

2014 - 2017

Computer Science

Graduation Science(Chem hons), Utkal University

2011 - 2014

Chemistry

CERTIFICATIONS

Azure Data Scientist Associate

2024

Data Science and Machine Learning by Microsoft

2019

HONORS & AWARDS

Entrepreneur Award, IBM

2023

Spot Appreciation Award, BIPROS

2019

Best Junior Employee, BIPROS

2018